32.1: Animals are multicellular, heterotrophic eukaryotes with tissues that develop from embryonic layers

Listing features shared by all animals is challenging, as there are exceptions to nearly every criterion we might select. When taken together, however, several characteristics of animals sufficiently describe the group for our discussion.

Nutritional Mode

Animals differ from both plants and fungi in their mode of nutrition. Plants are autotrophic eukaryotes capable of generating organic molecules through photosynthesis. Fungi are heterotrophs that grow on or near their food and that feed by absorption (often after they have released enzymes that digest the food outside their bodies). Unlike plants, animals cannot construct all of their own organic molecules, and so, in most cases, they ingest them—either by eating other living organisms or by eating nonliving organic material. But unlike fungi, most animals feed by ingesting their food and then using enzymes to digest it within their bodies.

Cell Structure and Specialization

Animals are eukaryotes, and like plants and most fungi, animals are multicellular. In contrast to plants and fungi, however, animals lack the structural support of cell walls. Instead, proteins external to the cell membrane provide structural support to animal cells and connect them to one another (see Figure 6.28). The most abundant of these proteins is collagen, which is not found in plants or fungi.

The cells of most animals are organized into **tissues**, groups of similar cells that act as a functional unit. For example, muscle tissue and nervous tissue are responsible for moving the body and conducting nerve impulses, respectively. The ability to move and conduct nerve impulses underlies many of the adaptations that differentiate animals from plants and fungi (which lack muscle and nerve cells). For this reason, muscle and nerve cells are central to the animal lifestyle.

Reproduction and Development

Most animals reproduce sexually, and the diploid stage usually dominates the life cycle. In the haploid stage, sperm and egg cells are produced directly by meiotic division, unlike what occurs in plants and fungi (see Figure 13.6). In most animal species, a small, flagellated sperm fertilizes a larger, nonmotile egg, forming a diploid zygote. The zygote then undergoes ****__cleavage__, a succession of mitotic cell divisions without cell growth between the divisions. During the development of most animals, cleavage leads to the formation of a multicellular embryonic stage called a **blastula**, which in many animals takes the form of a hollow ball **(Figure 32.2)**. Following this stage is the process of **gastrulation**, during which the layers of embryonic tissues that will develop into adult body parts are produced (see also Figure 47.8). The resulting developmental stage is called a **gastrula**.

Figure 32.2

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    Early embryonic development in animals.
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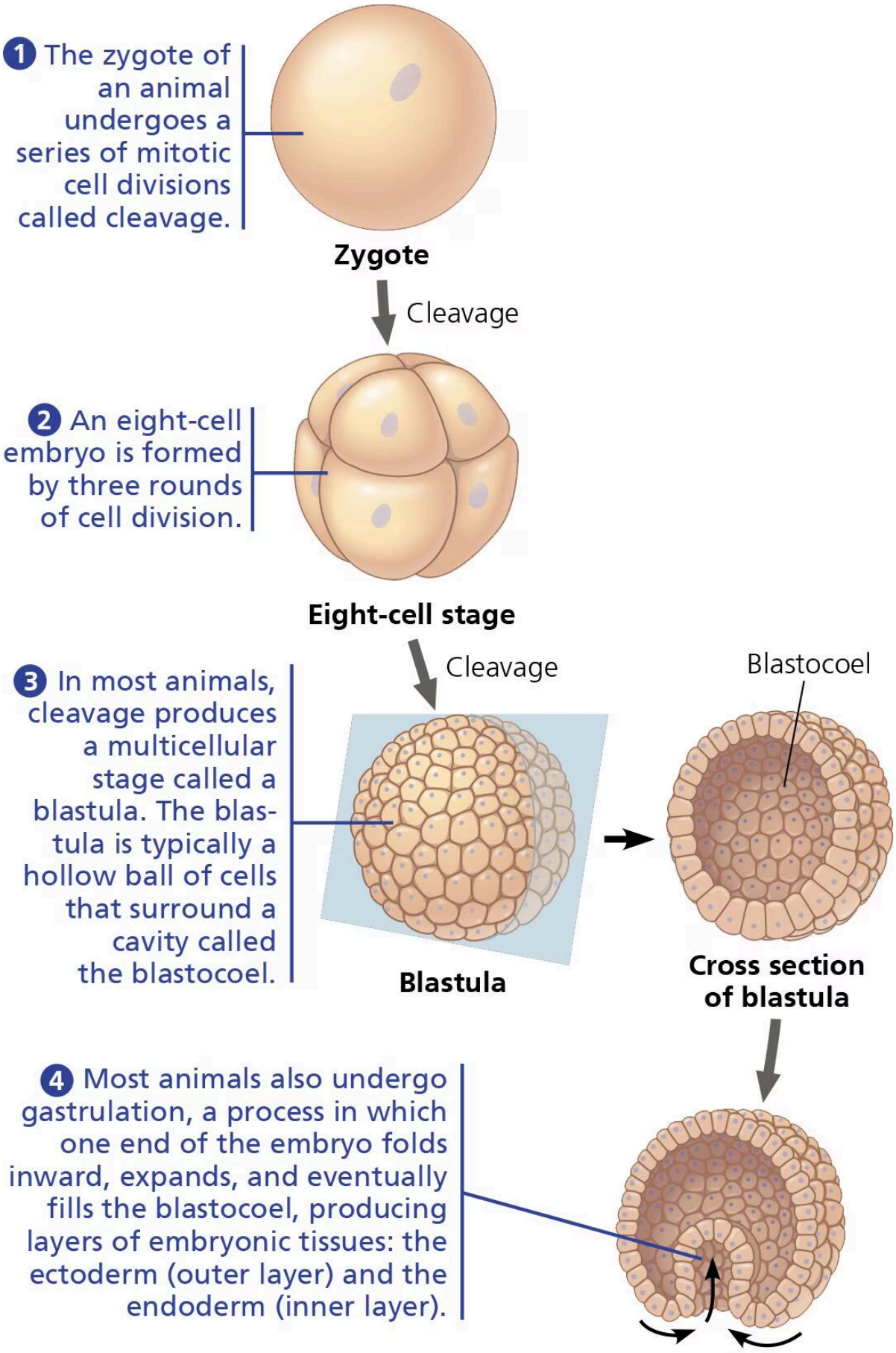
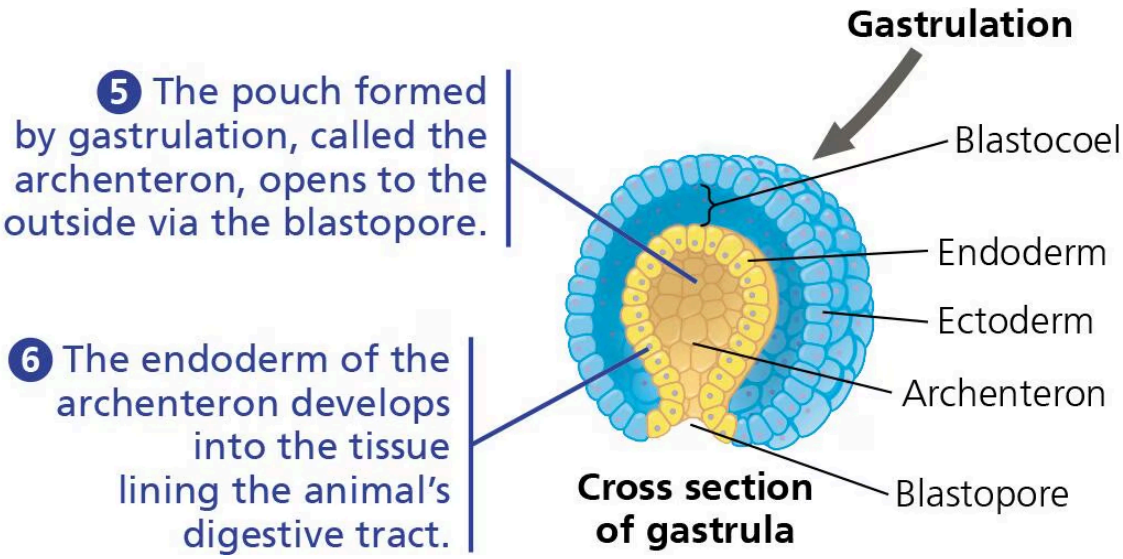

1 The zygote of an animal undergoes a series of mitotic cell divisions called cleavage.
Zygote
Cleavage
2 An eight-cell embryo is formed by three rounds of cell division.
Eight-cell stage
Cleavage
3 In most animals, cleavage produces a multicellular stage called a blastula. The blastula is typically a hollow ball of cells that surround a cavity called the blastocoel.
Blastocoel
Blastula
Cross section of blastula
4 Most animals also undergo gastrulation, a process in which one end of the embryo folds inward, expands, and eventually fills the blastocoel, producing layers of embryonic tissues: the ectoderm (outer layer) and the endoderm (inner layer).



Steps are as follows.

1.The zygote of an animal undergoes a series of mitotic cell divisions called cleavage.

2.An eight-cell embryo is formed by three rounds of cell division.

3.In most animals, cleavage produces a multicellular stage called a blastula. The blastula is typically a hollow ball of cells

4.that surround a cavity called the blastocoel.

5.Most animals also undergo

6.gastrulation, a process in which one end of the embryo folds inward, expands, and eventually fills the blastocoel, producing

7.layers of embryonic tissues: the ectoderm (outer layer) and the endoderm (inner layer).

8.The pouch formed by gastrulation, called the archenteron, opens to the outside via the blastopore.

9.The endoderm of the archenteron develops into the tissue lining the animal's digestive tract.

Although some animals, including humans, develop directly into adults, the life cycles of most animals include at least one larval stage. A **larva** is a sexually immature form of an animal that is morphologically distinct from the adult, usually eats different food, and may even have a different habitat than the adult, as in the case of the aquatic larva of a mosquito or dragonfly. Animal larvae eventually undergo **metamorphosis**, a developmental transformation that turns the animal into a juvenile that resembles an adult but is not yet sexually mature.

Though adult animals vary widely in morphology, the genes that control animal development are similar across a broad range of taxa. All animals have developmental genes that regulate the expression of other genes, and many of these regulatory genes contain sets of DNA sequences called *homeoboxes* (see Concept

21.6). In particular, most animals share a unique homeobox-containing family of genes, known as *Hox* genes. *Hox* genes play important roles in the development of animal embryos, controlling the expression of many other genes that influence morphology.